4 Performance evaluation parameters

The performance evaluation of a method is necessary to assure the validity of a method. This section lists out various performance measures that researchers have found useful for the quantitive evaluation of an image segmentation method [51]. Table 3 tabulates the formulation of various performance parameters.

1. Confusion matrix (CM): It is a widely used representation for the assessing the efficacy of a classification method [66]. However, it can be used to analyze the results of a clustering method too. The confusion matrix (CM) of size NxN represents that there are N classes (or clusters). In context of clustering, the number of correctly clustered patterns (true positive (TP), true negative (TN)) and wrongly clustered patterns (false positive (FP), false negative (FN)) can be easily identified from the confusion matrix. Table 4 illustrates an example of a typical confusion matrix for 2 clusters, i.e. positive and negative. In the table, TP and TN depict the number of data items which are correctly predicted in positive and negative clusters respectively. FP and FN correspond to the number of data items which are wrongly predicted as positive and negative clusters respectively. Based on Table 4, precision, recall and accuracy can also be computed using (18) – (20).

$$Precision = \frac{TP}{TP + FP} \tag{18}$$

$$Recall = \frac{TP}{TP + FN} \tag{19}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{20}$$

2. Intersection of Union (IoU): Intersection of Union is the ratio of the number of common pixels between X and Y to the total number of pixels in X and Y. Here, X and Y

Table 3 Various performance parameters for the quantitative evaluation of an image segmentation method [51]

Parameters	Formulation
Boundary Displacement Error (BDE)	$BDE = \begin{cases} \frac{u-v}{L-1}, & 0 < (u-v) \\ 0, & (u-v) < 0 \end{cases}$
Probability Rand Index (PRI)	$PRI = \frac{a+b}{a+b+c+d} = \frac{a+b}{L}$
Variation of Information (VoI)	$VoI = H(S_1) + H(S_2) - 2I(S_1, S_2)$
Global Consistency Error (GCE)	$GCE = \frac{1}{n}\left\{\sum_i E(S_1, S_2, p_i), \sum_i E(S_2, S_1, p_i)\right\}$
Structural Similarity Index (SSIM)	$SSIM = \frac{(2\times\bar{X}\times\bar{Y}+c_1)(2\times\sigma_{XY}+c_2)}{(\sigma_X^2+\sigma_Y^2+c_2)\times\lvert((\bar{X})^2+(\bar{Y})^2+c_1)}$
Feature Similarity Index (FSIM)	$FSIM = \frac{\sum_{x\in\Omega} S_L(x).PC_m(x)}{\sum_{x\in\Omega} PC_m(x)}$
Root Mean Squared Error (RMSE)	$\text{RMSE} = \sqrt{\text{MSE}(\hat{\theta})} = \sqrt{\text{E}((\hat{\theta}-\theta)^2)}$
Peak Signal to Noise Ratio (PSNR)	$PSNR = 10log_{10}\frac{(2^b-1)^2}{\sqrt{MSE}}$
Normalized Cross-Correlation (NCC)	$\text{NCC} = \frac{1}{n}\sum_{x,y}\frac{1}{\sigma_X\sigma_Y}\left(X(x,y)-\overline{X}\right)\left(Y(x,y)-\overline{Y}\right)$
Average Difference (AD)	$AD = \frac{1}{MN}\sum_{x=1}^{M}\sum_{y=1}^{N}(X(x,y)-Y(x,y))$
Maximum Difference (MD)	$MD = max \mid X(x,y) - Y(x,y) \mid$
Normalized Absolute Error (NAE)	$NAE = \frac{\sum_{x=1}^{M}\sum_{y=1}^{N}\lvert X(x,y)-Y(x,y)\rvert}{\sum_{x=1}^{M}\sum_{y=1}^{N}\lvert X(x,y)\rvert}$

Table 4 Confusion matrix

	True Positive	True Negative
Predicted Positive	TP	FP
Predicted Negative	FN	TN

correspond to the segmented image and ground truth respectively. The formulation of IoU is depicted in (21).

$$IoU = \frac{|X \cap Y|}{|X| + |Y|} \tag{21}$$

3. Dice-Coefficient (DC): Dice-Coefficient is defined as twice the number of common pixels divided by total number of pixels in X and Y, where X corresponds to the segmented image and Y is the ground truth. DC is mathematically defined as (22).

$$DC = \frac{2|X \cap Y|}{|X| + |Y|} \tag{22}$$

4. Boundary Displacement Error (BDE): This parameter computes the average boundary pixels displacement error between two segmented images as depicted in (23). The error of one boundary pixel is defined as it's distance from the closest pixel in the other boundary image.

$$\mu_{LA}(u, v) = \begin{cases} \frac{u-v}{L-1} & 0 < u - v \\ 0 & u - v < 0 \end{cases} \tag{23}$$

5. Probability Rand Index (PRI): Probability Rand Index finds labelling consistency between the segmented image and its ground truth. It counts such fraction of pairs of pixels and average the result across all ground truths of a given image as shown in (24).

$$R = \frac{a + b}{a + b + c + d} = \frac{a + b}{n} \tag{24}$$

6. Variation of Information (VOI): Variation of Information as shown in (25) computes the randomness in one segmentation in terms of the distance from given segmentation.

$$VOI(X; Y) = H(X) + H(Y) - 2I(X, Y) \tag{25}$$

7. Global Consistency Error (GCE): A refinement in one segmentation over the other is depicted by the value of GCE as given in (26). If two segmentations are related this way then they are considered as consistent, i.e. both can represent the same natural image segmentation at different scales.

$$GCE = \frac{1}{n}\left\{\sum_i E(s_1, s_2, p_i), \sum_i E(s_2, s_1, p_i)\right\} \tag{26}$$

8. Structural Similarity Index (SSIM): Structural Similarity Index measures the similarity between two images by taking initial uncompressed or distortion-free image as the reference and computed as (27). It incorporates important perceptual phenomena such as luminance masking and contrast masking.

$$SSIM = \frac{(2 \times \bar{x} \times \bar{y} + c_1)(2 \times \sigma_{xy} + c_2)}{(\sigma_x^2 + \sigma_{y)}^2) \times ((\overline{x})^2 + (\overline{y})^2 + c_1)} \tag{27}$$

9. Feature Similarity Index (FSIM): Feature Similarity Index is a quality score that uses the phase congruency (PC), which is a dimensionless measure and shows the significance of a local structure. It is calculated by (28).

$$FSIM = \frac{\sum_{x \in \Omega} S_L(x).PC_m(x)}{\sum_{x \in \Omega} PC_m(x)} \tag{28}$$

10. Root Mean squared error (RMSE): The root-mean-squared error computes the difference between sample value predicted by a model or an estimator and actual value. The formulation is shown in (29).

$$\mathrm{RMSE}(\hat{\theta}) = \sqrt{\mathrm{MSE}(\hat{\theta})} = \sqrt{\mathrm{E}((\hat{\theta} - \theta)^2)} \tag{29}$$

11. Peak Signal to Noise Ratio (PSNR in dB): Peak signal-to-noise ratio is defined as the ratio between the maximum possible power of a signal and the power of corrupting noise and is calculated using (30). In general, a higher value of PSNR represents high quality reconstruction and is defined via MSE.

$$PSNR = 10log_{10}\frac{(2^n - 1)^2}{\sqrt{MSE}} \tag{30}$$

12. Normalized Cross-Correlation (NCC): Normalized cross-correlation is used for template matching where images are first normalized due to lighting and exposure conditions. It is calculated by subtracting the mean of original and segmented images from the corresponding images, and divided by their standard deviations. Let $t(x, y)$ is the segmented image of $f(x, y)$, then NCC is calculated by (31).

$$\frac{1}{n}\sum_{x,y}\frac{1}{\sigma_f \sigma_t}\left(f(x, y) - \overline{f}\right)\left(t(x, y) - \overline{t}\right) \tag{31}$$

13. Average Difference (AD): This parameter represents the average difference between the pixel values and is computed by (32).

$$AD = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}(x(i, j) - y(i, j)) \tag{32}$$

14. Maximum Difference (MD): This parameter finds the maximum of error signal by taking the difference between original image and segmented image and defined in (33).

$$MD = max \mid x(i, j) - y(i, j) \mid \tag{33}$$

15. Normalized Absolute Error (NAE): The normalized absolute difference between the original and corresponding segmented image gives NAE and is calculated by (34).

$$NAE = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} \mid x(i, j) - y(i, j) \mid}{\sum_{i=1}^{M}\sum_{j=1}^{N} \mid x(i, j) \mid} \tag{34}$$

In the above mentioned parameters, IoU, DC, SSIM, FSIM, PRI, PSNR, and NCC show better segmentation on high values. A high value indicates that the segmented image is more close to the ground truth. To measure the same, these parameters compute values by considering the region of intersection between the segmented image and the ground truth which corresponds to matching the number of similar pixels. On the contrary, other indices prefer lower values for better segmentation as these measures compute error between the segmented image and the ground truth and aim at reducing the difference between them.

5 Image segmentation benchmark datasets

The considered dataset is the key for fair analysis of an image segmentation method. The validation of segmentation methods against benchmark datasets tests its performance against the challenges posed by the considered dataset. Generally, a benchmark dataset consists of a variety of images that varies in number of different aspects. Some of the common challenges which segmentation methods need to handle are illumination variation, intra-class variation, and background complexity. This paper lists some of the popular benchmark datasets in Table 5 which are publicly available for image segmentation task and are briefed below.

- **Aberystwyth Leaf Evaluation Dataset:** It is dataset of timelapse images of Arabidopsis thaliana (Arabidopsis) plant to perform leaf-level segmentation. It consists of original arabidopsis plant images with 56 annotated ground truth images [1].
- **ADE20K:** It is a scene parsing segmentation dataset with around 22K hierarchically segmented images. For each image, there is a mask to segment objects and different parts of the objects [3].

Table 5 Benchmark datasets for image segmentation [23]

S.No.	Dataset	Purpose	Links
1	Aberystwyth Leaf Evaluation Dataset	Segmentation of Leaf	[1]
2	ADE20K Dataset	Segmentation of object and different parts of the objects	[3]
3	Berkeley Segmentation Dataset and Benchmark (BSDS)	Segmentation of objects	[74, 78]
4	Brain MRI Dataset	Segmentation of FLAIR abnormalitiy	[14]
5	CAD120 Affordance Dataset	Segmentation of Affordance of objects	[15]
6	CoVID-19 CT-images Segmentation Dataset	Segmentation of infected region	[21]
7	Crack Detection Dataset	Segmentation of cracks	[5]
8	Daimler Pedestrian Segmentation Benchmark	Segmentation of pedestrian	[24]
9	Epithelium Segmentation Dataset	Segmentation of epithelium region	[80]
10	EVIMO Dataset	Motion segmentation	[30]
11	Liver Tumor Segmentation Dataset	Segmentation of Liver and Liver Tumor	[39]
12	Materials in Context Dataset	Material segmentation	[57]
13	Nuclei Segmentation Dataset	Segmentation of breast cancer nuclei	[79]
14	Objects with Thin and Elongated parts	Segmentation of objects with thin and elongated parts	[20]
15	OpenSurfaces Dataset	Surface segmentation	[56]
16	Oxford-IIIT Pet	Segmentation of pet animals	[82]
17	PetroSurf3D Dataset	Segmentation of petroglyphs	[35]
18	Segmentation Evaluation Database	Segmentation of single or two objects	[67]
19	Sky Dataset	Segmentation of sky	[71]
20	TB-roses-v1 Dataset	Segmentation of rose stems	[26]
21	Tsinghua Road Markings Dataset	Segmentation of road markings	[63]

- **Berkeley Segmentation Dataset and Benchmark (BSDS):** This is a benchmark dataset for evaluation of image segmentation method. It is a collection of 12K manually-labelled segmentations performed on 1K images from the Corel dataset by 30 human subjects. There are two versions of this dataset. BSDS300 [74] and BSDS500 [78] consist of 300 and 500 number of images respectively.
- **Brain MRI dataset:** It is segmentation dataset of MRI images along with manual fluid-attenuated inversion recovery (FLAIR) abnormality segmentation masks [14].
- **CAD120 Affordance Dataset:** This segmentation dataset provides binary masks for each the affordance in the RGB image captured from the Cornell Activity Dataset (CAD) 120. The affordance annotation is performed in the context of human [15].
- **CoVID-19 CT-images Segmentation Dataset:** This dataset consists of 100 CT-scan images of more than 40 CoVID-19 patients along with segmented ground truths. The segmented images were annotated by an expert radiologist [21].
- **Crack Detection Dataset:** It consists of five datasets where each dataset consists of pavement images and associated segmented images for crack detection [5].
- **Daimler Pedestrian Segmentation Benchmark:** This dataset contains pedestrian images along with corresponding segmented ground truth to perform pedestrian segmentation [24].
- **Epithelium Segmentation dataset:** There are 42 ER+ breast cancer images scanned at the magnification of 20x. The ground truth contains manually annotated epithelium region by an expert pathologist [80].
- **EVIMO Dataset:** This is a motion segmentation dataset along with egomotion estimation and tracking captured through event camera. It provides pixel-level masks for the motion segmentation [30].
- **Liver Tumor Segmentation (LITS):** This is a public available benchmark dataset provided by CodeLab for segmentation of liver tumor. It contains 3D-CT scans of liver along with segmented liver and liver tumor [39].
- **Materials in Context (MINC):** This dataset is intended for the segmentation and recognition of materials from the image. It consists of point annotations for 23 different types of materials [57].
- **Nuclei Segmentation dataset:** This dataset contains 143 ER+ breast cancer images scanned at 40x. In total, there are around 12,000 nuclei segmented manually. [79]
- **Objects with Thin and Elongated Parts:** It is a database of three datasets where each image consists of objects having thin and elongated parts. In total, there are 280 images of birds and insects with corresponding ground truth images [20].
- **OpenSurfaces:** This dataset includes tens of thousands of segmented surfaces. The dataset is constructed from interior photographs and annotated with information like material type, texture, and contextual [56].
- **Oxford-IIIT Pet Dataset:** This dataset consists of pet images with their pixel-level trimap segmentation. There are 37 categories of pets and each category contains 200 images [82].
- **PetroSurf3D Dataset:** This is a dataset which contributes in the area of rock art. There are total 26 high-resolution 3D-scans images with pixel-wise labelling for the segmentation of petroglyphs [35].
- **Segmentation Evaluation Database:** This dataset contains 200 gray-level images along with manually generated ground-truth images by three human subjects. Each image consists of either one or two objects in the foreground and the ground truth images are annotated into two or three classes [67].